Cisco 7945G

Home telephony

Asterisk in k3s + Home Assistant

Two SIP lines, color display and native PoE

Cisco 7945G: model and inventory

2

SIP lines

Native 802.3af

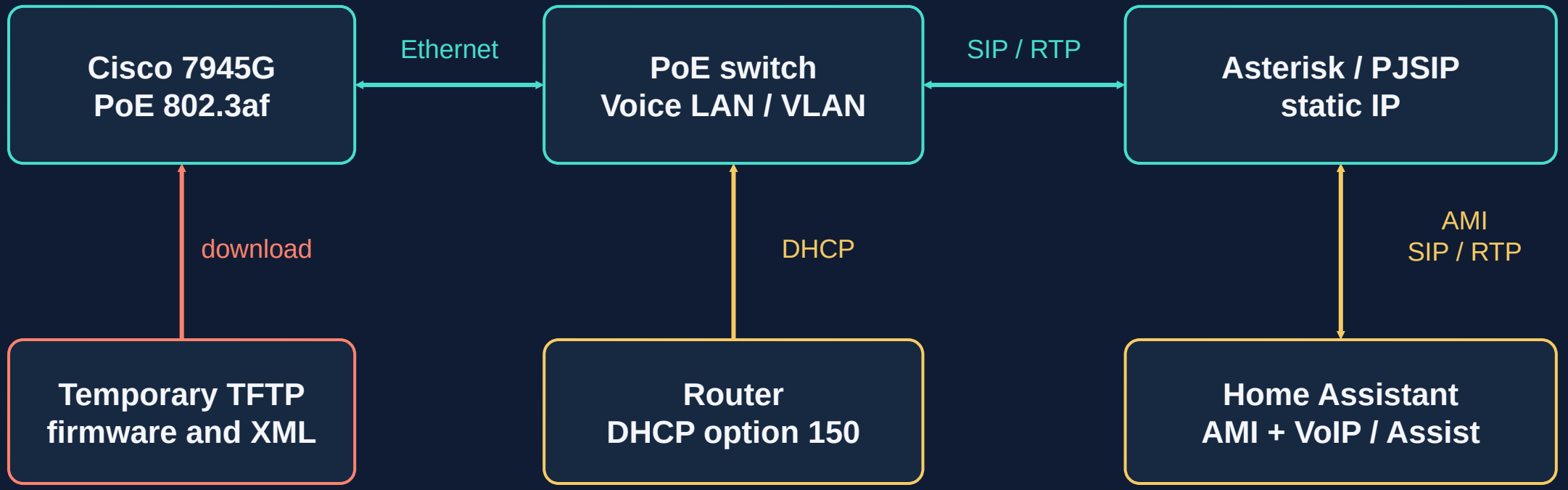
7945_7965 SIP firmware family

A phone running SCCP needs SIP firmware.
This model does not need MPP conversion.

Before starting

Record PID, VID/HW, MAC, App Load,
Boot Load, IP and TFTP for every phone.

Cisco, Asterisk and Home Assistant



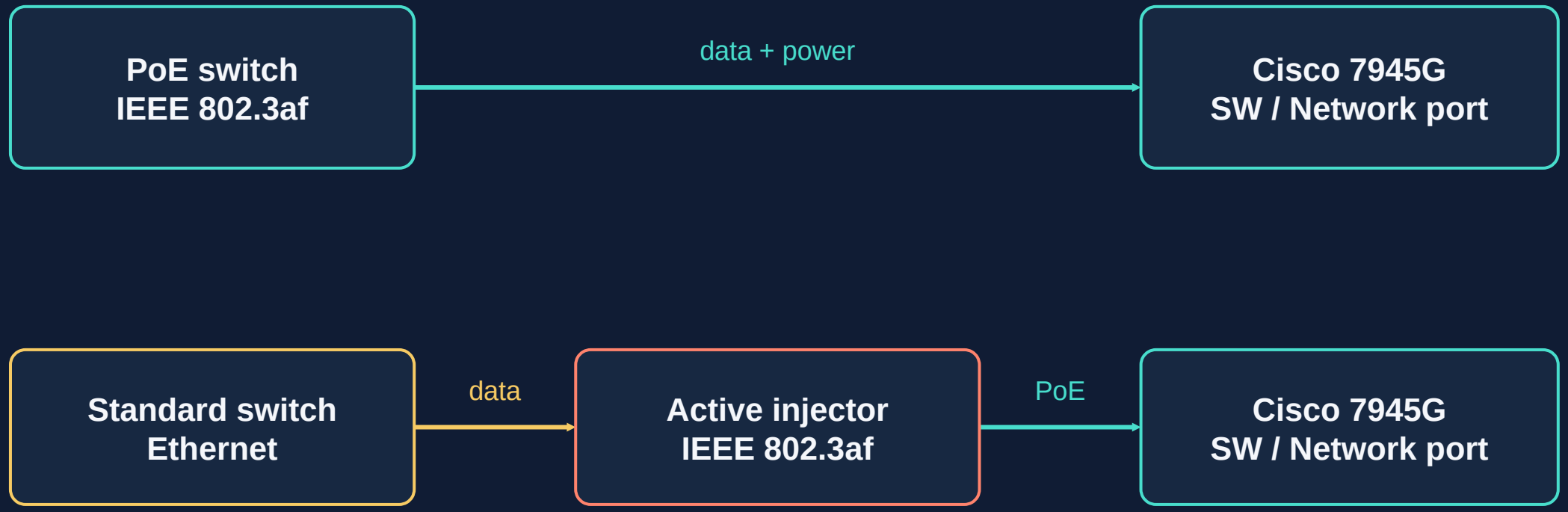
Asterisk uses hostNetwork and one replica. HA opens the AMI connection to the PBX.
TFTP runs only during provisioning. SIP, RTP and AMI stay on the LAN / VPN.

Signaling and audio ports

5060	TCP / UDP	SIP: Cisco TCP, HA UDP
10000–10199	UDP on PBX	RTP / audio
5038	TCP on PBX	AMI: HA / gateway /32
69 + 20000–20031	UDP on PBX	Temporary TFTP

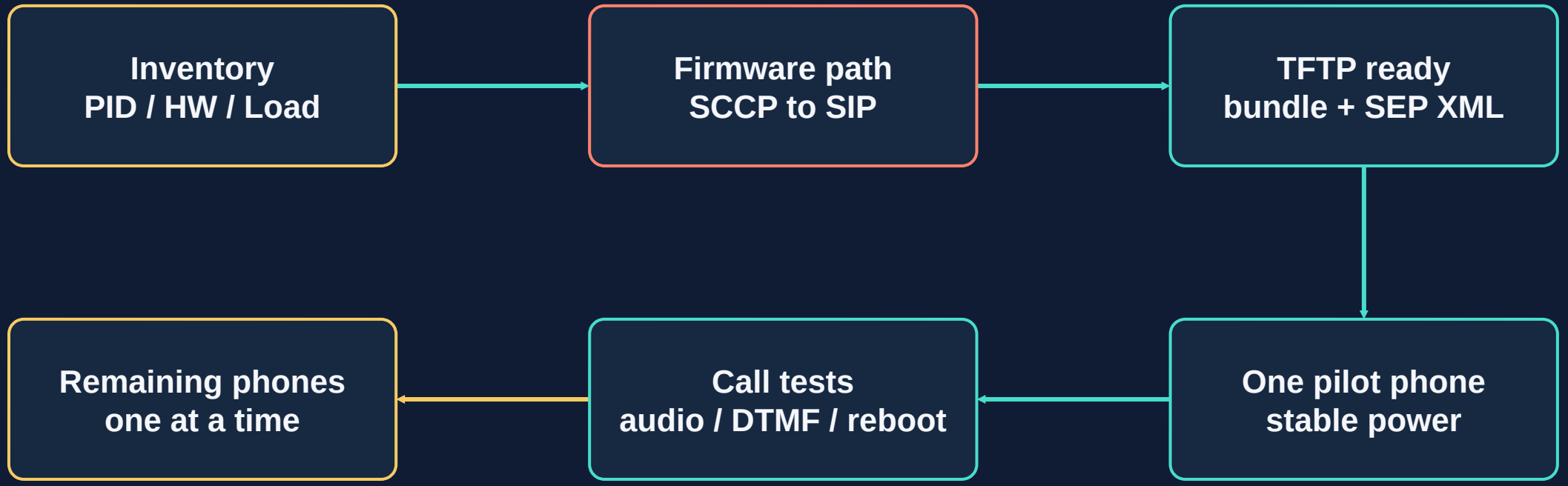
HA uses dynamic RTP / RTCP ports. Restrict firewall rules to the required source IPs.

7945G has built-in PoE support



A passive 24 V injector is unsuitable. The UPS also needs to power the switch, router and k3s node.

One pilot phone before the full fleet



HW 13.0+ requires at least 9.3(1)SR3. The upgrade path depends on the current load. Prepare TFTP and recovery before a reset. Keep power connected during flash writes.

TFTP preparation before restart

Complete official firmware bundle

.loads and every referenced binary file
with the original filenames and letter case

SEP<MAC>.cnf.xml

SIP server, extension and a unique password
Load ID from the selected 7945_7965 firmware branch

Use DHCP option 150 on the home router or Alternate TFTP on the phone.

Asterisk: independent SIP lines

201 / 202

one phone

One endpoint, auth and AOR per line
A unique password for each line

G.711 ulaw / alaw
direct_media=no keeps RTP on Asterisk

600 echo, 601 tone, 620 ring group. Extension 630 requires Assist.

AMI: status and call control

Home Assistant connects to PBX:5038

Status, connected line and DTMF
Actual entity IDs from your HA instance

Place calls and hang up a specific channel

asterisk.send_action: Originate / Hangup
The DTMF example sends a harmless notification

Use a separate secret and a /32 ACL. The system permission allows administrative actions.

630: voice assistant on the phone



First call: enable Allow calls and select a pipeline in HA.
Then test speech recognition, the spoken reply and hangup.

SIP From retains the phone extension. HA uses dynamic UDP ports for RTP / RTCP.
A call on the pilot phone verifies audio. Pod readiness checks server components.

Deployment and commissioning

01 **Server bootstrap**

Images, static IP, firewall, Asterisk and Opus

02 **Verified pilot phone**

PID / HW / Load / MAC, SIP profile, temporary TFTP

03 **Home Assistant setup**

AMI /32, secret, actual entity IDs and VoIP pipeline

04 **Acceptance and rollout**

Two phones, a call to 630, then the remaining phones

Acceptance tests on real phones

Telephony

201 and 203: cross-phone audio, hold / transfer

Home Assistant

States, Originate, Hangup on the correct Channel, DTMF

Assist 630

Speech recognition, spoken reply and hangup

Repeat after phone, PBX and HA restarts. Readiness does not verify audio.

Operations and recovery

Private backups outside Git

Inventory, SIP / AMI secrets and Asterisk data
HA config / .storage, entity mappings and image versions

One node is a single point of failure

UPS for network equipment and the k3s node
A recovery plan that preserves the LAN IP

Exclude raw DTMF and connected-line data from Recorder. Review existing history separately.

Server readiness and phone acceptance

Server preparation can come first

Images, configuration, AMI and Opus
Tests verify the server components

A pilot verifies the phone experience

Phone inventory, a two-party call and Assist
Add the remaining phones after acceptance

Physical acceptance is a separate step. Keep the results in the private deployment log.

Ten Cisco identities, twenty lines

PHONE 1	PHONE 2	PHONE 3	PHONE 4	PHONE 5
201 / 202	**203 / 204**	**205 / 206**	**207 / 208**	**210 / 211**
PHONE 6	PHONE 7	PHONE 8	PHONE 9	PHONE 10
212 / 213	**214 / 215**	**216 / 217**	**218 / 219**	**220 / 221**

620: one primary line per Cisco device

201 · 203 · 205 · 207 · 210 · 212 · 214 · 216 · 218 · 220

Keep each device’s identity and credentials when unplugged. Online status follows the connected phones.

Home Assistant phone controls

290

controls across ten phones

Call controls

Line, dial, answer and exact-channel hangup

Personal settings

PBX DND and optional normal-call auto-answer

Schedules

Timers and weekdays with a Stop alert control

Phone screen

Authenticated text and approved images

Manual timer dismissal and HA Stop passed. A scheduled alarm rang beyond 90 seconds.

Radio and scheduled alerts

RING

Until you stop it

Answer / reject
or press Stop alert

No auto-answer
No after-answer audio

701	Matryoshka Radio
702	Echo online
703	RADIO BOB! Rock
704	RADIO BOB! Metal
705	RADIO BOB! Gothic
706	SomaFM PopTron
707	Radyo Viva / Turkish

For 708–710: handset down, speaker off, enter all digits, then press Dial.

Free SIP account status

SIP2SIP	Echo answered, RTP loss observed
Callcentric	Registered, outbound review required
Linphone SIP	803 answered, two-way RTP observed
FWD	Stopped at mandatory Verification Fee
Google Voice	No official free Asterisk SIP trunk

Answered calls and RTP verified. Human listening and DTMF menu tests remain pending.

Mac calling and children’s radio

209

Zoiper5 · Mac 209

Own SIP identity
Internal calls and services
Excluded from ring group 620

708	**Toddler music**
709	**Lullabies**
710	**Children’s songs**

Official Deti FM channels

Mac 209 echo: RTP and human listening passed. All three stations are deployed; 708 was heard.